



SHRI SHANKARACHARYA INSTITUTE OF PROFESSIONAL
MANAGEMENT & TECHNOLOGY, RAIPUR

(Approved by AICTE-New Delhi, Affiliated to CSVTU-Bhilai)



IEEE

Student Branch - SSIPMT

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2025

NAVONMESH 25

A NATIONAL LEVEL HACKATHON

On Real Time Challenges of Raipur Nagar Nigam (CG)

TEAM NAME : Code bringers

THEME : Water bodies and river conservation

PS CATEGORY : Software & Hardware



IDEA TITLE

Our Solution



Water Bodies and River Conservation

Water bodies are natural or artificial collections of water, such as rivers, lakes, ponds, wetlands, reservoirs and oceans, which are essential for life, biodiversity, agriculture, industry, and climate regulation. River conservation is a focused set of activities and policies aimed at protecting, restoring, and maintaining rivers and their ecosystems, ensuring they remain healthy and sustainable for present and future generations.

E - Soil

- E- soil (electronic soil).
- Water conservation.
- Urban agriculture.
- Sustainable materials.
- Feeding a growing population sustainable

Real time prediction

- Detecting the water & soil quality.
- intelligent dashboard.
- It is use backend, frontend, Database, etc.



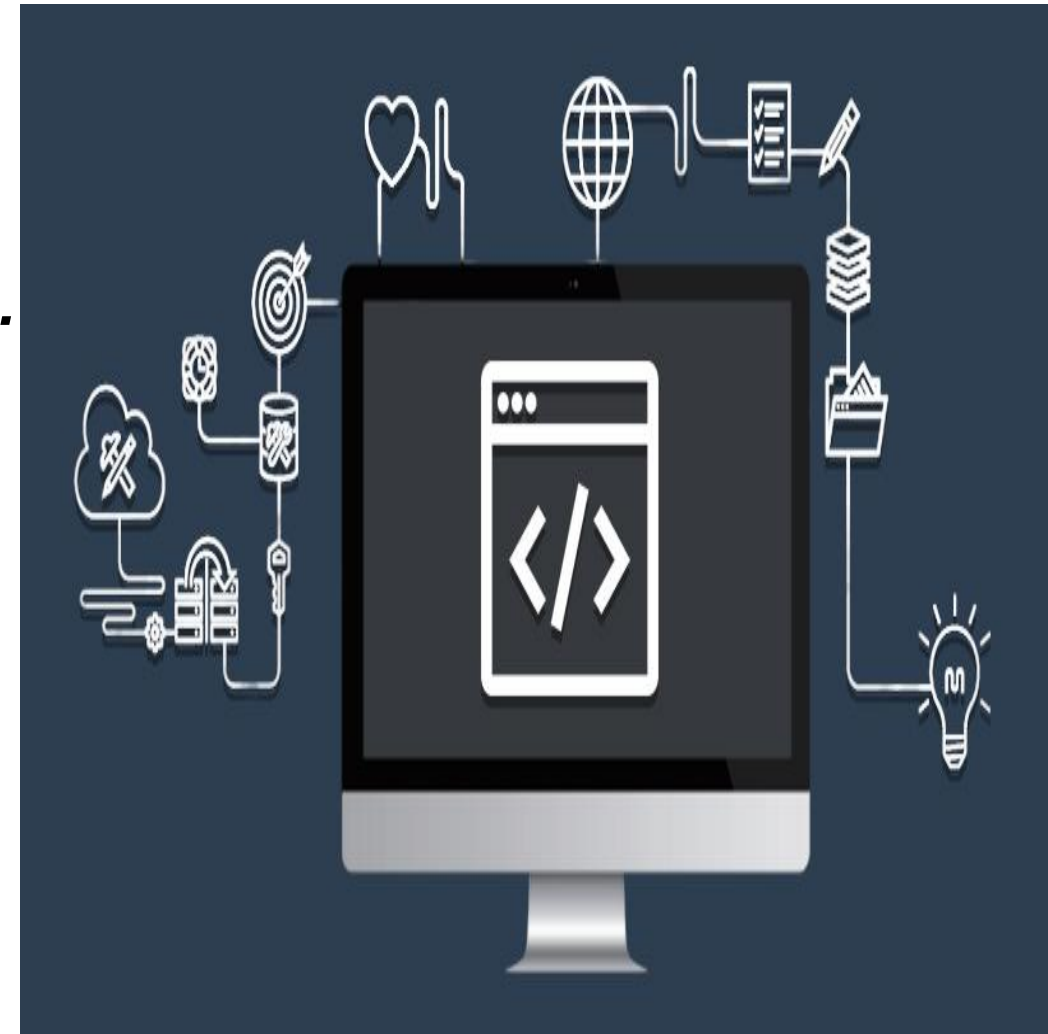
TECHNICAL APPROACH

Technology stack

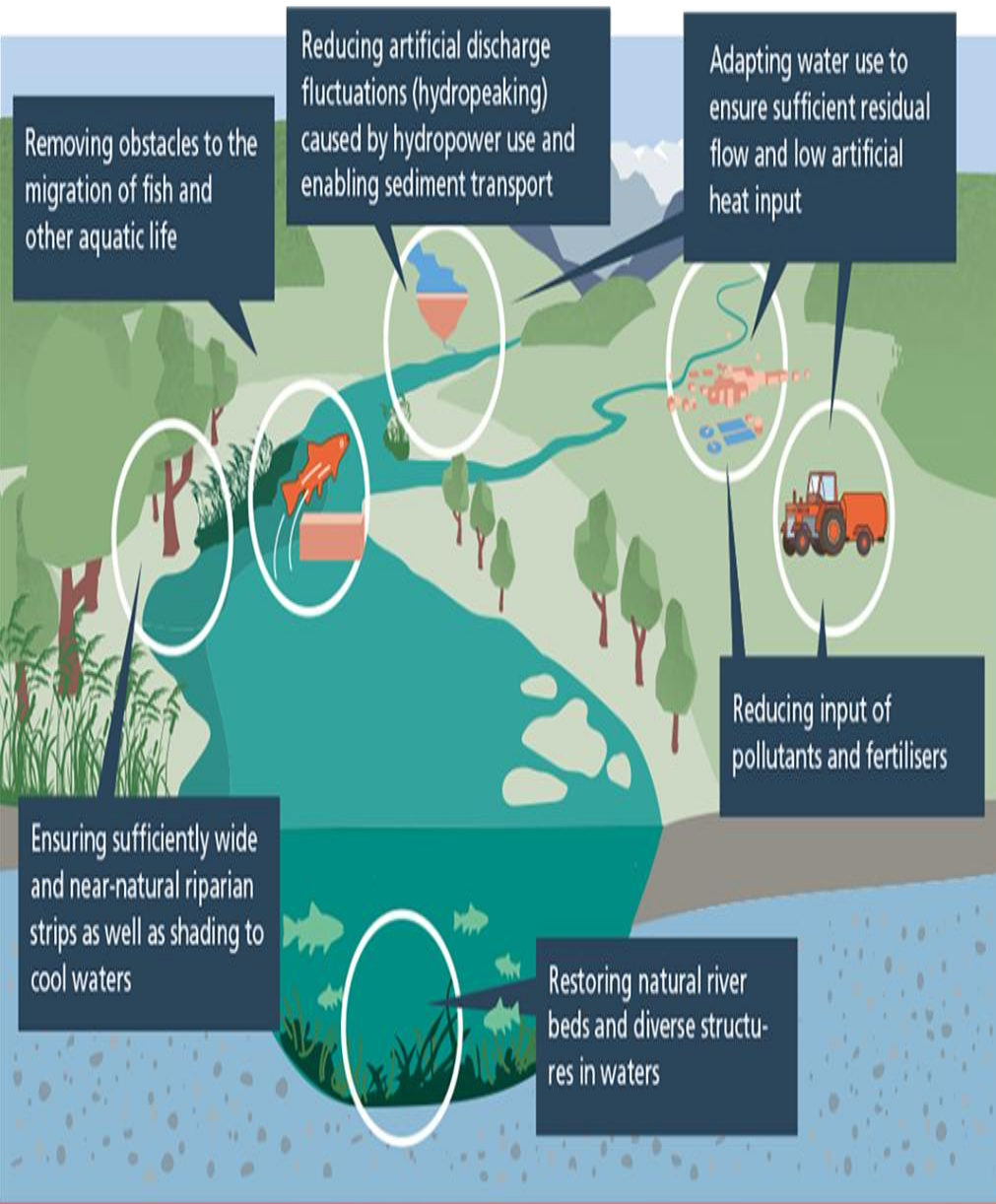
- **Frontend** : Html , JavaScript , CSS .
- **Backend** : python, Node .j s , Flask , Advance AI/ML.
- **Database** : Mongo DB , SQL.
- **Hardware** : Hardware tools .

Key point

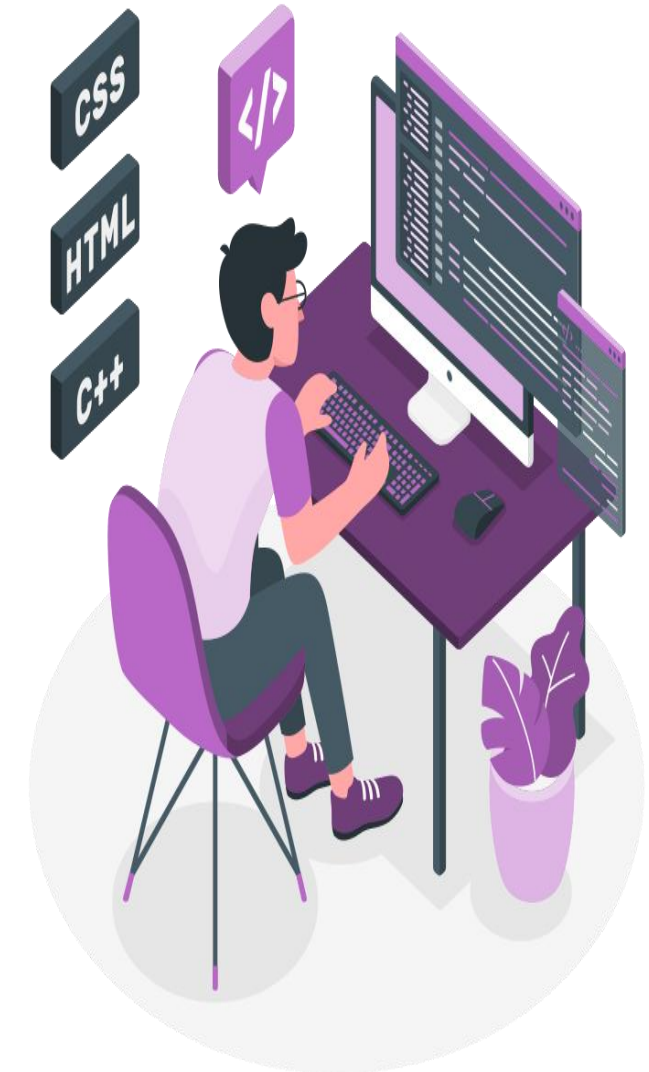
- Model train for data sheet.
- Real time detection.



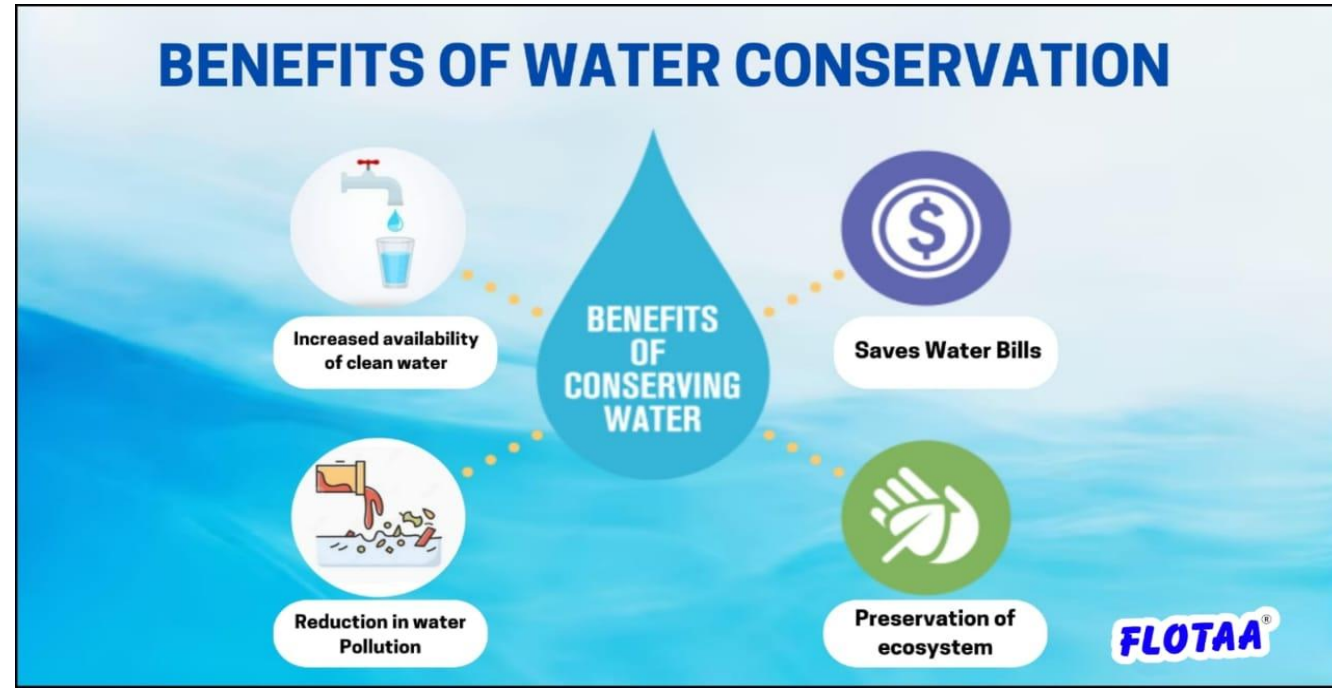
FEASIBILITY AND VIABILITY



- The project solution E-soil is developing technique which can very expensive.
- Detecting the water and soil quality- what kind of element are present in water & soil , & it is harmful or not.
- The properties are which are good for water & soil and which are bad for water and soil.
- In order to make project more accurate it requires the precise detailing of element.



IMPACT AND BENEFITS



- It help farmer to boost their crop production.
- It help government to clean rivers.
- The project gives clear knowledge about quality of water.
- It specify the element which is higher & lower then normal limit.
- It tells uses to add element which is lacked in water.



RESEARCH AND REFERENCES

Key research focus

E-Soil/Dielectric Sensing :- supports the methodology for non- invasive soil moisture and nutrient measurement using electrical properties, avoiding traditional, slow methods.

AI/ML in Water Management :- Provides the framework for **predictive analytics** and real-time anomaly detection for contamination (e.g., predicting nutrient spikes or pollution events) using algorithms.

Urban Agriculture & Conservation :- Justifies the need for **data-driven decision-making** to optimize water and fertilizer use, supporting the "Urban Agriculture.

References

A Non-Invasive Soil Moisture Sensing System Electronic Architecture :- Explores the electronic architecture and feasibility of **non-invasive** soil moisture detection, directly supporting the 'E-Soil' concept.
<https://www.mdpi.com/1424-8220/20/21/614>

Advancements in dielectric soil moisture sensor Calibration :- A comprehensive review of the methods and techniques required for accurate **dielectric soil moisture sensors**, crucial for data accuracy. Link-
<https://www.ars.usda.gov/research/publications/publication/?seqNo115=406421>

IOT BASED WATER AND SOIL QUALITY MONITORING SYSTEM :-Details the use of multiple sensors (pH, EC, Moisture) and **IOT/Cloud** technologies for real-time monitoring, relevant to your Technical Approach . link –
<https://www.semanticscholar.org/paper/IOT-BASED-WATER-AND-SOIL-QUALITY-MONITORING-SYSTEM-Loganathan-Dr.E.Mohan/3e8bc46e4ab59fa1e4994c6fda0447191dd4901e>

Real-Time IOT-Enabled Water Management for Rooftop Urban Agriculture :- **Water Management for Rooftop Urban Agriculture** Provides a practical example of integrating low-cost **IOT sensors** and microcontrollers for water management in urban farming environments.

Link- <https://www.mdpi.com/2673-4583/10/1/34>